



CWW-Det: A Context-aware Wavelet-enhanced Detector for SAR Ship Detection

Zihao Wang¹, Jinwei Wang¹, Zhenguang Gou¹, Xinyu Wang¹, Yuzhao Tang¹, Yongjin Zhao¹

1. College of Physics and Electronic Information Yantai University

Abstract

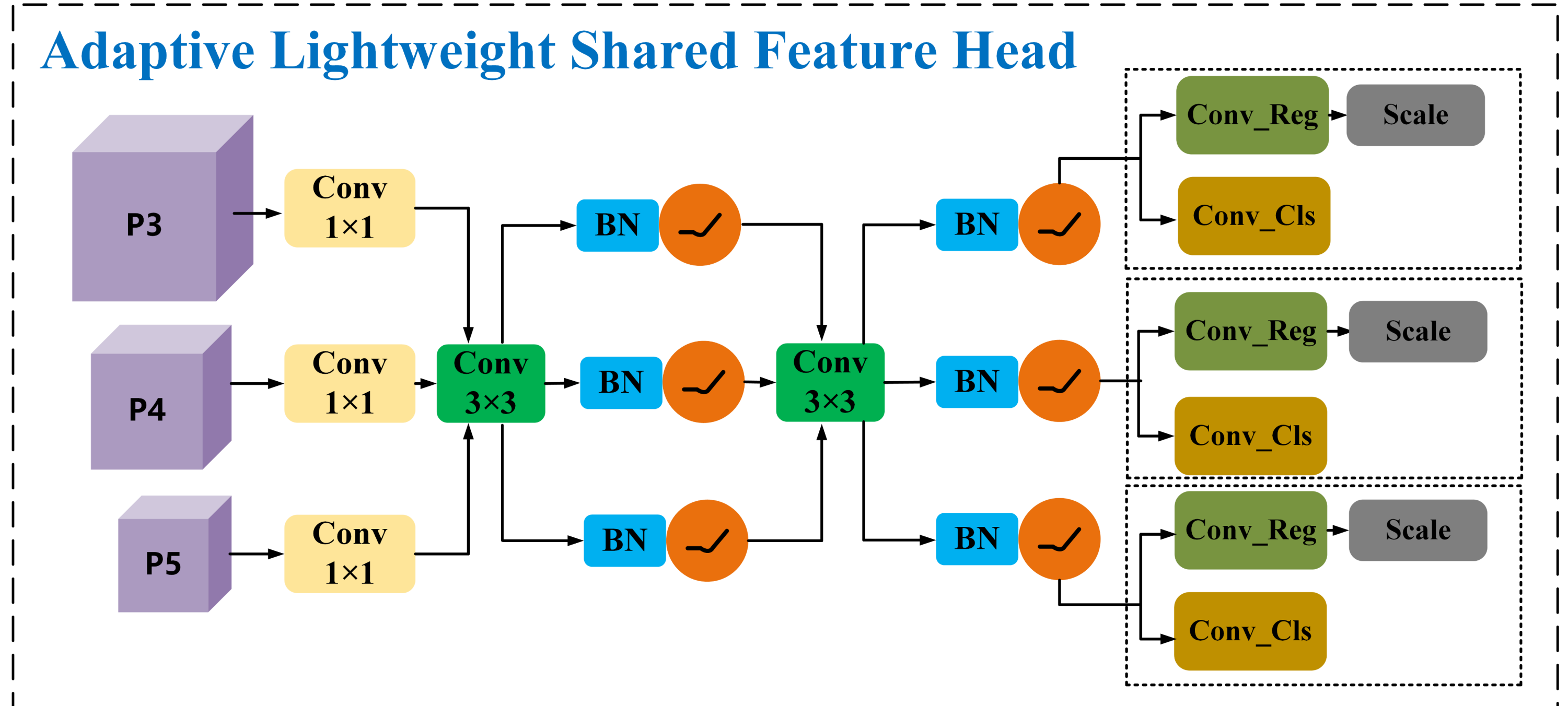
Synthetic Aperture Radar (SAR) imagery enables all-weather maritime monitoring, but ship detection remains challenging due to low contrast, complex backgrounds, multi-scale variations, and partial occlusions. To address these issues, this paper proposes a Context-aware Wavelet-enhanced Detector (CWW-Det), a lightweight SAR ship detection network. The network consists of three main components: the ContextWave-Backbone (CWB), which integrates a Joint Context Refinement Block (JCRB) for global context modeling and wavelet pooling for frequency-preserving multi-scale feature extraction; the Flexible Feature Pyramid Network (Flexi-FPN), which enhances feature fusion through refined upsampling and improved convolutions; and the Adaptive Lightweight Shared Feature Head (ALSF-Head), which reduces computational cost while improving prediction accuracy via shared convolutions and scale-independent normalization. Experiments on the public HRSID dataset and the SAR-Ship-Dataset demonstrate that CWW-Det achieves superior detection accuracy and robustness while maintaining low complexity, providing an efficient and generalisable solution for SAR ship detection.

Introduction

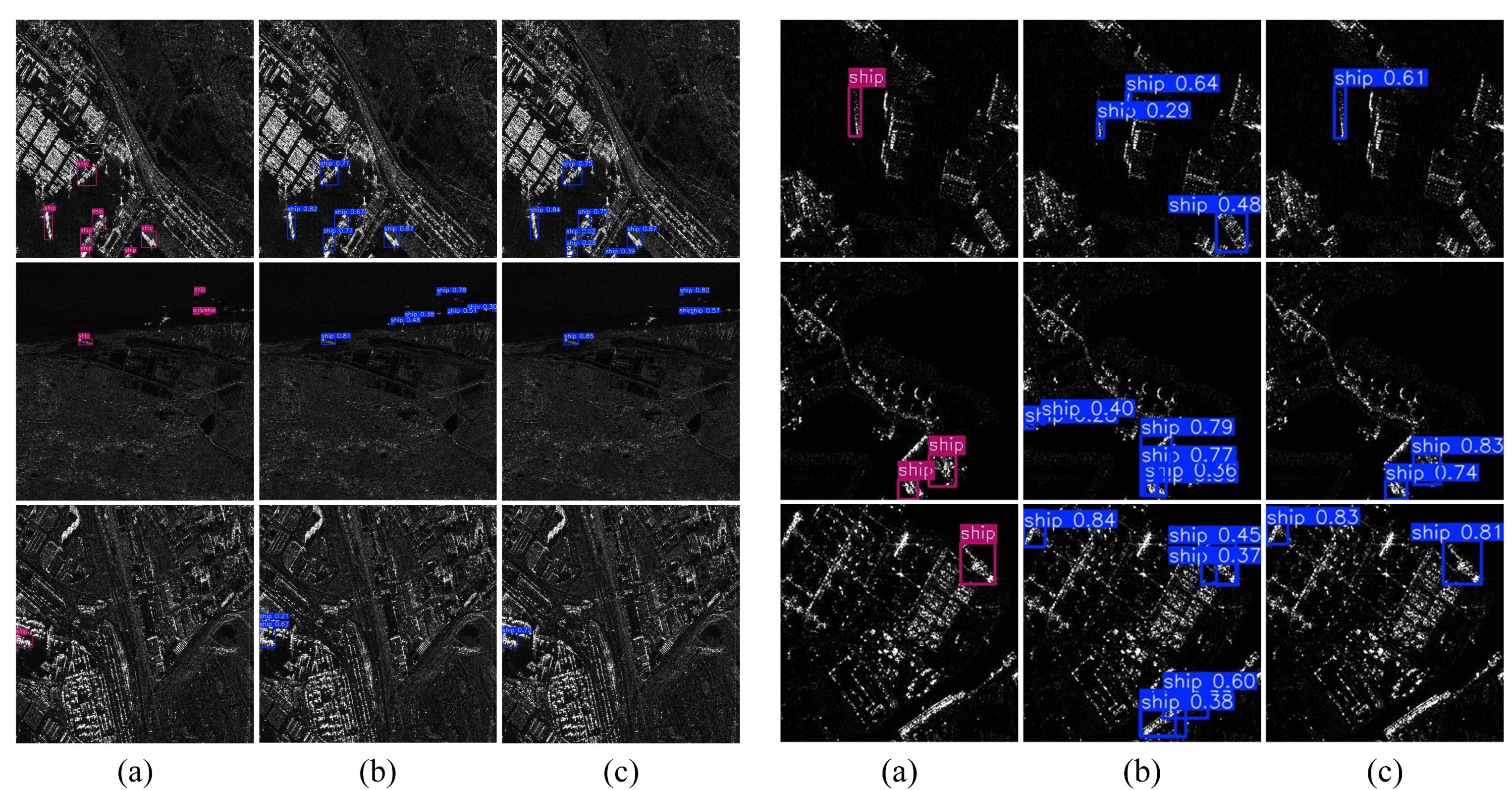
CWW-Det is a lightweight SAR ship detection network designed to handle low contrast, complex backgrounds, and multi-scale targets in SAR imagery. Its core components include:

1. ContextWave-Backbone (CWB): Combines a Joint Context Refinement Block (JCRB) for global context modeling with Wavelet Pooling to preserve multi-frequency structural and edge details.
2. Uses Soft Nearest Interpolation (SNI) for refined upsampling and GSConv-V2 to enhance receptive fields and feature discrimination under cluttered backgrounds.
3. Adaptive Lightweight Shared Feature Head (ALSF-Head): Applies shared convolutions to reduce parameters and introduces Scale-Independent Batch Normalization (SBN) to adapt to different feature scales and noise characteristics.
4. Output: Fuses multi-scale features and predicts ship locations efficiently with improved accuracy and reduced computational cost.

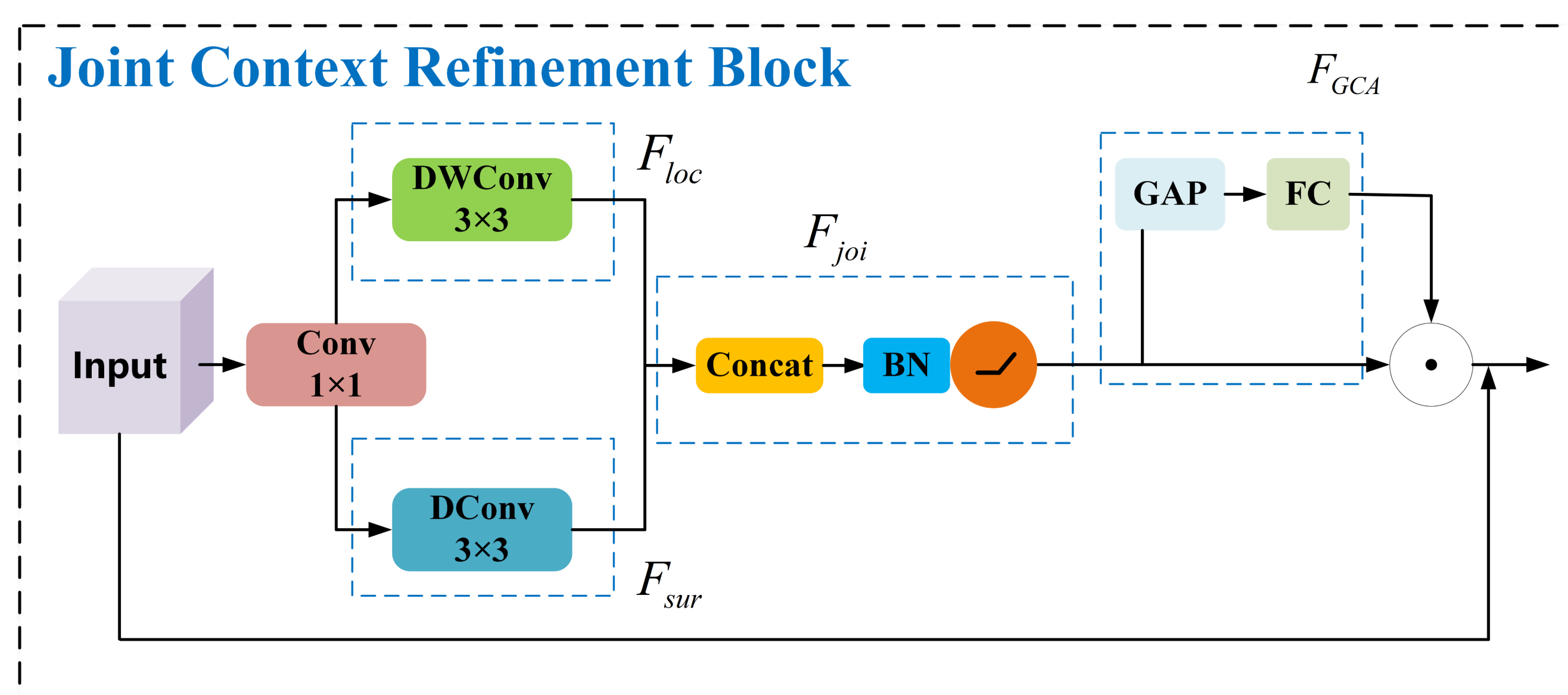
ALSF-Head



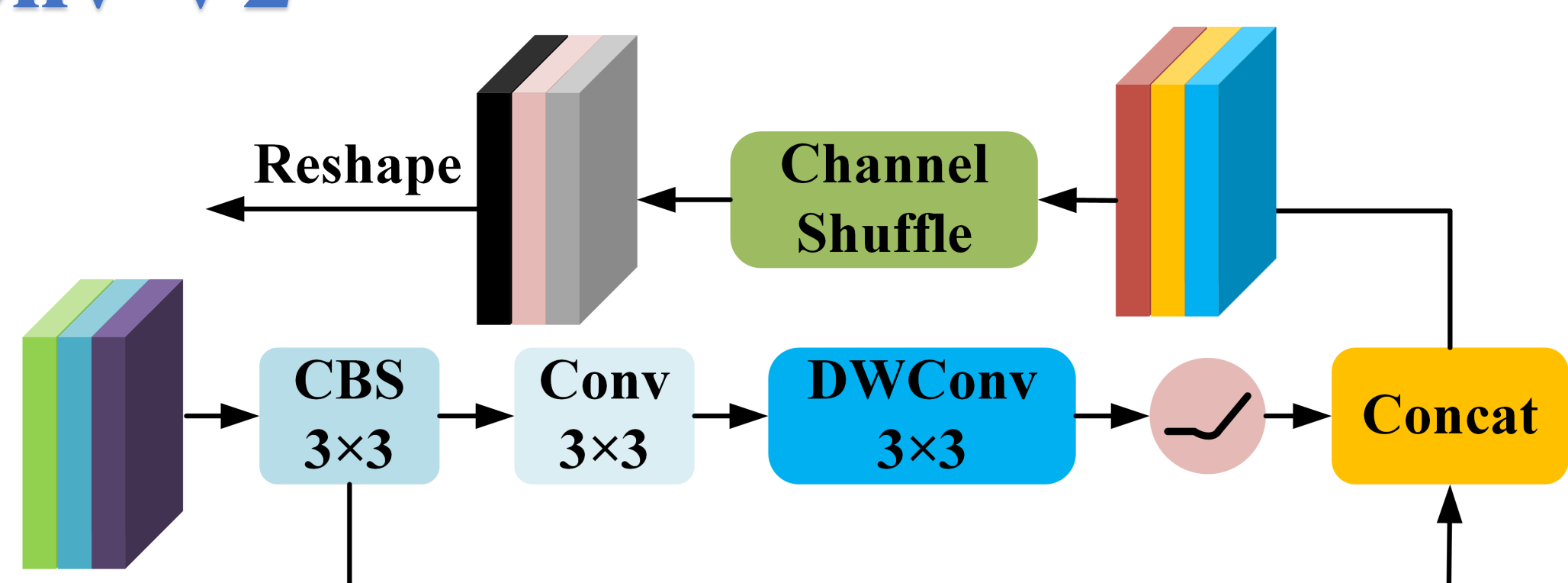
Result



JCRB



GSConv-V2



Results of SAR Ship Detection in Comparative Experiments. In the figure: (a) Original SAR image, (b) Baseline detector results, (c) Proposed CWW-Det results.

Left: HRSID dataset; Right: SAR-Ship-Dataset.

TABLE II
THE COMPARATIVE EXPERIMENT ON THE HRSID DATASET.

Models	Params(M)	FLOPs(G)	mAP50%	R%	FPS
YOLO11n	2.58	6.3	91.2	82.5	136.2
YOLO12n	2.56	6.3	90.1	80.4	109.8
PEG-Net	-	-	90.5	-	-
LDSS-Net	2.0	8.7	91.7	91.2	-
CWW-Det	1.46	4.3	92.3	84.1	154.3